A proof-of-concept analysis of a semiautomatic literature-review tool for abstract and title screening

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Abstract: Title and abstract screening consumes approximately 20% of the time spent on systematic reviews, yet existing large language model (LLM) solutions remain mostly proprietary, expensive, and cloud-dependent.This study presents a minimal, open-source, single-LLM pipeline, accompanied by a local web application that makes the tool accessible to researchers without programming experience, designed for semi-automated screening on consumer hardware with low computational cost. We validated the pipeline across three distinct systematic reviews: RV1 (2,840 articles, very low prevalence), RV2 (131 articles, high prevalence), and RV3 (1,597 articles, fuzzy criteria). Two prompt architectures (single-message vs. system+user) and two low-cost models: (1)cloud-based DeepSeek-V4-Flash and (2)local qwen3:8B (via Ollama) were tested in a 3×2×2 full factorial design. Across configurations, recall against the original included articles ranged from 77% to 100%, with negative predictive values ≥99% in most runs. Workload reduction varied from 22% (RV2, high prevalence) to 99% (RV1, very low prevalence). F1 scores fell between 0.06 and 0.31, as expected given extreme class imbalance. Prompt structure had no consistent effect while domain and model choice were more influential. The local qwen3 model matched or outperformed DeepSeek in two of the three domains, while eliminating costs and privacy risks. We conclude that a single-LLM low-cost pipeline is feasible for semi-automated screening, provided a human-in-the-loop validates all exclusions. The choice between local and cloud models should be guided by domain sensitivity, budget, and privacy requirements.

1. Introduction

Systematic reviews of biomedical literature plays a central role in consolidating scientific knowledge and supporting evidence-based clinical practice.[1,2] However, conducting systematic reviews is a notoriously time-consuming, complex, and costly process.[1,3,4] It is estimated that a conventional review takes on average 67.3 weeks from conception to publication, with an indexed total cost of around 141,000 USD.[1] Approximately 20% of this time and financial effort is absorbed exclusively by the initial title and abstract screening (T&A) phase.[1] Partial automation of this step therefore represents an opportunity for significant efficiency gains without compromising methodological rigour.[1,3,4]

Although the emergence of large language models (LLMs) such as GPT, Claude, or Gemini has opened real prospects for partial automation of this process, demonstrating sensitivities between 56,2% and 98,8% and substantial workload reductions (34% to 37,5%), their practical application still faces severe methodological barriers.[1,4] The most recent literature unanimously points to three central challenges that limit the routine adoption of these tools: (i) prompt variability as the main source of performance metric fluctuation;[1] (ii) the modest inter-human reference agreement, with a Cohen's κ of only 0.645, which sets a realistic ceiling for machine-human agreement;[5] and (iii) the need to prioritise the Negative Predictive Value (NPV) and workload reduction, since the primary goal of semi-automated screening is to safely exclude irrelevant articles, while final inclusion always remains dependent on human confirmation.[4]

Despite these advances, a critical gap persists: the scarcity of screening tools that are simultaneously accessible, transparent, and adaptable to different research teams.[4] Most approaches proposed in the literature resort to complex multi-model architectures, closed-source commercial APIs, or workflows that assume budgets incompatible with the reality of many research groups and institutions.[2,4] This dependency not only imposes recurring costs but also introduces privacy and confidentiality risks that hinder integration into regulated clinical or academic settings.[4,6] Therefore, simplified, open-source, locally executable solutions that combine competitive performance with economic sustainability and data integrity are lacking.[2,4,6]

To address this gap, the present work describes the development and validation of a minimalist semi-automated tool for article screening. Differentiating itself from existing solutions, the proposed pipeline is based on a single-LLM architecture, without complex multi-model voting mechanisms, and is designed under the principles of strict reproducibility, open source, and full execution on consumer hardware. The tool was evaluated on three systematic reviews with distinct characteristics, crossing two prompt architectures (V1, unified, and V2, with system message) and two low-cost production models: DeepSeek-V4-Flash (cloud) and qwen3:8B (local, via Ollama).The two central methodological questions guiding the study are: (i) evaluate the impact of prompt structure on screening effectiveness when comparing the unified approach (V1) with system-user separation (V2); and (ii) to what extent can local open-weight models or very low-cost cloud models achieve comparable performance and workload reduction without sacrificing the safety of excluding irrelevant articles or introducing associated privacy risks. The article is structured as follows: Section 2 details the materials and methods; Section 3 presents the results; Section 4 discusses the implications and limitations; and Section 5 concludes with recommendations for practice and future research.

2. Materials and Methods

The developed pipeline is organized into nine sequential layers, conceptually divided into two main phases, as illustrated in Figure 1:

Phase 1 - Screening and production pipeline (Layers 1, 2, 3, 4, 5, 7A), dedicated to reference import, pre-processing, development, and general execution of the tool; and Phase 2 - Validation,outputs,evaluation and reporting (Layers 6, 7B, 8, 9), focused on the review interface, metric extraction, and experimental validation of the tool.

The tool was developed entirely in Python, prioritising reproducibility, transparency and execution on consumer hardware. The tool was developed entirely in Python, prioritising reproducibility, transparency and execution on consumer hardware. The source code is available at [https://github.com/goncalofebra/llm-abstract-screening] under an MIT licence.

2.1 Tool Architecture and Implementation

The pipeline design was inspired by the methodological approaches of Li et al. (2024) [2] for unified prompt structuring, and by Dennstädt et al. (2024) [3] for optimising system messages in modern conversational models. The workflow begins with the definition of the systematic review's conceptual parameters and culminates in the classification call via an Application Programming Interface (API). Specifically, following the model of Li et al. (2024) [2], we adopted the strategy of formulating instructions that require a strictly binary response format (”yes” or ”no”) for each of the evaluated inclusion and exclusion criteria. This requirement of single-word answers per criterion eliminates ambiguity in the language model's decisions and facilitates the automated extraction of the final classification. In turn, the influence of Dennstädt et al. (2024) [3] is reflected in the modular architecture of the message sent to the model, which is built by concatenating independent text blocks: the base instruction, the publication title, the abstract, and the relevant criteria.

2.1.1) Layer 1: Systematic Review Inputs

The input layer (Figure 1, Layer 1) gathers three types of conceptual information that define the review for screening purposes: (a) conceptual parameters, namely the title, short citation, thematic scope, and eligible study types; (b) inclusion and exclusion criteria, rigorously extracted from the PRISMA protocol or the methodological section of the original review; and (c) the document source, which may consist of a search query run on reference databases such as PubMed, Scopus, or Web of Science, or a citation export file (CSV/XLSX) containing titles and abstracts. These input data were channelled directly to the execution of the preparation scripts (Layer 2).

2.1.2) Layer 2: Preparation Scripts

The construction of the prompt files is carried out by four independent utility scripts (Figure 1, Layer 2), run once per review. This organic separation ensures auditability, since the content sent to the model is traceable via executable commands, and provides flexibility when modifying criteria. Among these modules, the make_baseprompt.py file generates the generic instruction, dictating the topic, constraints, and the strict binary output requirement. make_criteria.py formats the selection criteria, while make_system.py concatenates the instruction and criteria into a single file, a process essential for supporting the V2 architecture of the pipeline. Finally, title_abstract_corpus.py converts XLSX/CSV exports into the native TXT input format. These preparation scripts merely serve to mitigate inconsistencies and facilitate the construction of prompts used by the LLMs.

2.1.3) Layer 3: Generated Artifacts

The preparation scripts produce two sets of artifacts (Figure 1, Layer 3). In the pipeline/prompts/ lder, the TXT prompt files are stored: baseprompt_<RV>.txt, which describes the conceptual parameters of the review, namely the title, short citation, scope, and eligible study types; criteria_<RV>.txt, with the inclusion and exclusion criteria explicitly stated; and title_abstract_<RV>.txt, in which each record follows the fixed format Title: / Abstract: / Label Included:. The Label Included field serves for later evaluation and can be ignored when using the tool solely for its intended purpose without validation in a real screening context.

2.1.4) Layer 4: LLM Screening

The computational core of the pipeline implements a screening engine that queries a large language model for each record. For a given article, the engine assembles a message containing the base instruction, the inclusion/exclusion criteria, and the title-abstract pair. The model is then required to produce a strictly binary response (“yes” or “no”) for each criterion, which a subsequent parser consolidates into a final classification (“include”/“exclude”). The message structure is modular, allowing different prompt architectures (e.g., single user message or separation of system and user roles). The specific architectures tested during validation are described in detail in section 2.2.

2.1.5) Layer 5: LLM API Call and Model Transition

The inference call to the computational model was abstracted by an integrating argument designed to support OpenAI, DeepSeek, and Ollama infrastructures in an agnostic manner. The initial exploratory phase used GPT-4o, accessed remotely via its official API (available at https://platform.openai.com). This model served as the

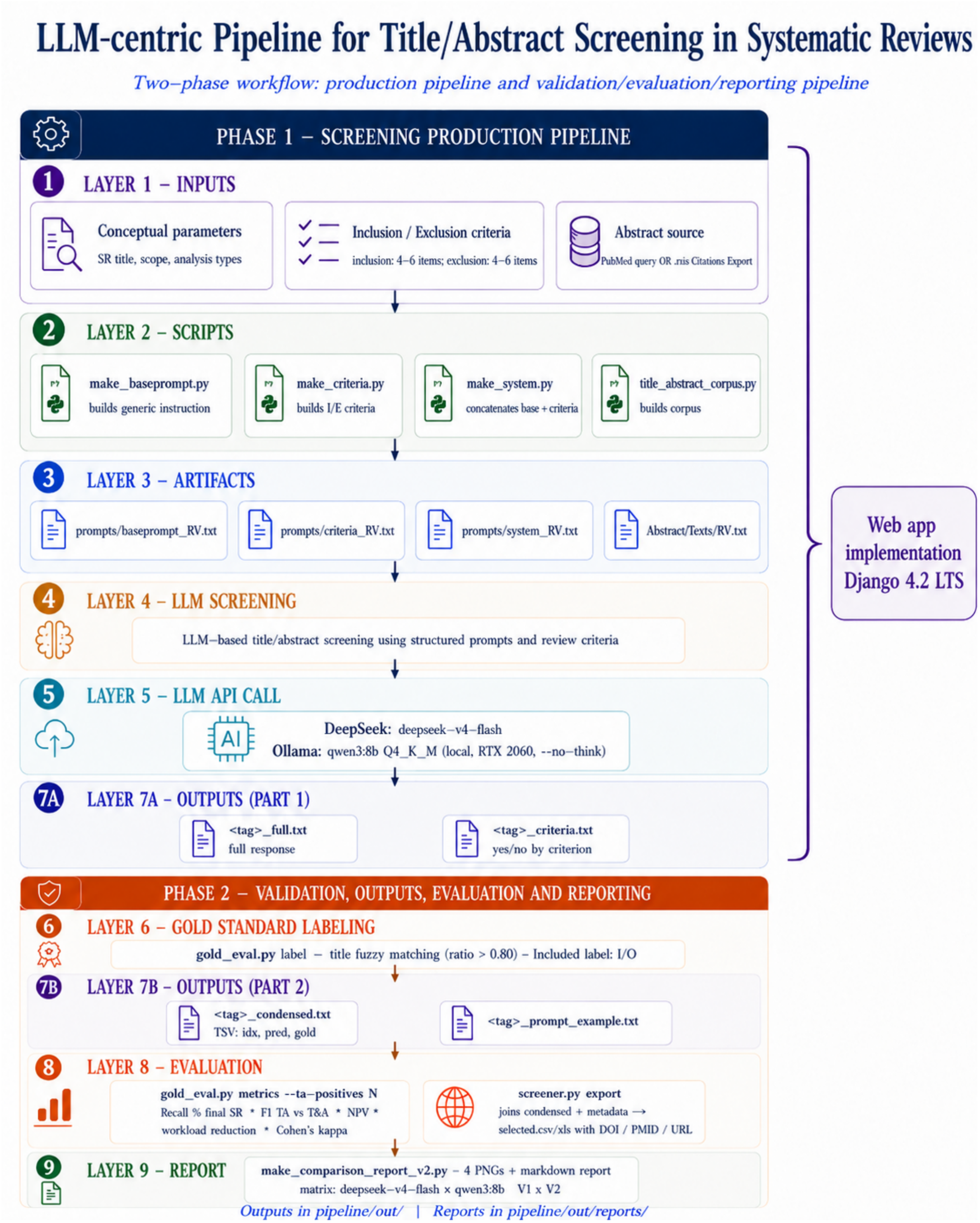


Figure 1 - LLM-centric pipeline for title and abstract screening in systematic reviews. Adapted from Li et al. 2024 and Dennstädt et al. 2024.

upper reference for initial prompt calibration and was later kept exclusively as an isolated contextual baseline for the evaluation of the third systematic review and cost assessment optimization(supplementary material figure 1). However, for the final matrix, we adopted the cloud-based DeepSeek-V4-Flash model.

Developed by DeepSeek, this remotely accessed model demonstrated exceptional analytical quality and was integrated into the pipeline using the native OpenAI client library (version 1.40.0) in Python, requiring only the replacement of the base address to the platform's endpoint (https://api.deepseek.com) and the injection of the respective cryptographic authentication key. The native hybrid reasoning behaviour of this model was inhibited,

2.1.6) Layer 7A: Outputs During the screening stage, the pipeline generated two granular output files per run, each capturing the model's response at a different level of detail and identified by a run specific tag (for example, RV3_full.txt). The _full.txt file stored the complete, unedited response returned by the LLM for every record, together with that record's index, title, abstract and gold label, providing a full audit trail that allowed any individual decision to be traced back to the exact text the model produced. The _criteria.txt file decomposed each decision into its constituent parts, recording the binary yes/no answer for every eligibility criterion in the order in which the criteria were defined, alongside the overall inclusion decision. This per criterion breakdown supported error analysis by revealing which specific criterion drove a given inclusion or exclusion. Both files were written incrementally, one record at a time and with an immediate flush to disk, so that a run could be inspected while still in progress and recovered if interrupted, and both were stored in the pipeline/out/ directory.

2.2 Empirical Validation and Tool Optimisation

2.2.1 Study Design and Validation Protocol

The validation of the tool was based on a rigorous evaluation matrix designed to measure the effectiveness of the pipeline in screening articles. This phase relied on the three validation datasets (described in Section 2.2.3), using each systematic review as the gold standard against which the tool's accuracy was assessed.

The operational protocol followed a human-in-the-loop philosophy. The tool acted as a classifier assigning a relevance score to each article based on the probability of meeting the inclusion criteria. The usage protocol defined that the final user (the human User) would only interact with articles classified as "Relevant" or "Uncertain", while all articles classified with high confidence as "Irrelevant" were automatically excluded from the manual review. To quantify the success of this approach, we used three core metrics: (1) Workload reduction, calculated as the proportion of the entire corpus that the pipeline could automatically exclude, reflecting real time savings since the user was spared from reading the majority of articles deemed irrelevant by the model.(2) Sensitivity (recall), defined as the ratio of correctly identified relevant articles to the total number of articles actually included in the final review. This served as the most critical safety metric, ensuring that the system did not miss eligible studies. (3) Precision, assessed the model's ability to avoid incorrectly labelling irrelevant articles as relevant, thereby optimising user effort by reducing the need to verify false positives. Since this was a retrospective validation study, using the three previously published systematic reviews as the gold standard, the language model had no access to human decisions during inference (blinded evaluation). The authors also did not adjust the prompts based on the validation results before the final execution.

2.2.2) Layer 6: Gold Standard Labeling

The gold_eval.py script marked, the field of each record, the articles actually included in the original review. The matching between the list of included titles and the corpus records was performed by fuzzy matching (difflib.SequenceMatcher) with a similarity threshold of 0.80, robust to minor differences in punctuation or formatting between the published title and the PubMed index. The result constituted the gold standard against which all LLM predictions were evaluated. This module was used exclusively for experimental validation purposes and was not required for the operational functioning of the pipeline.

2.2.3) Layer 7B: Outputs for validation

For validation purposes, the pipeline generated two additional output files per run. The _condensed.txt file was a tab-separated values (TSV) file containing three columns: the record index (idx), the model's prediction (pred), and the gold standard label (gold). This file facilitated the rapid computation of performance metrics. The _prompt_example.txt file stored a representative example of the exact prompt sent to the LLM for a given run, supporting auditing and reproducibility. Both files were stored in the pipeline/out/ directory alongside the standard outputs (Layer 6).

2.2.4) Layer 8: Evaluation, Export, and T&A Level F1 Metric

The central evaluation module computed the confusion matrix and derived the fundamental statistical metrics from it. These included sensitivity (recall), precision (positive predictive value), F1 score, specificity, negative predictive value (NPV), and human workload reduction. Additionally, we calculated Cohen's κ coefficient to assess the classificatory agreement between the computational model and the published gold standard, noting that this value should not be directly compared with classical inter-rater agreement metrics. The most critical methodological decision was the adoption of a dual reading of the gold standard. Because evaluating only the finally included articles artificially inflates false negatives (exclusion often results from information hidden in the full text rather than the abstract), the pipeline adopted two levels of analysis.

The first level focused on final sensitivity, calculated as Recall = TP / (TP + FN), to verify whether the model inadvertently discarded crucial empirical evidence. The second level focused on sensitivity and precision at the strict title-and-abstract reading level, reflecting the exact decision point that the algorithm aimed to automate, producing a methodologically equitable F1 value. Statistical software: All analyses were performed in Python 3.9.12 (Anaconda). scikit-learn 1.6.1 (sklearn.metrics) was used to build the confusion matrix (confusion_matrix) and to compute Cohen's κ (cohen_kappa_score) and accuracy (accuracy_score). The remaining metrics (sensitivity/recall, precision, F1, specificity, NPV, and workload reduction) were computed directly from the confusion-matrix counts (TP, FP, TN, FN) using their standard definitions, rather than through scikit-learn functions.

2.2.5) Layer 9: Comparative Report Generation

In the final layer, the script make_comparison_report.py aggregated the results from the multiple runs and consolidated them into a structured report accompanied by supporting graphical visualisations. The report included four PNG images and a Markdown document comparing the performance of DeepSeek-V4-Flash against qwen3:8B across both prompt architectures (V1 and V2).

2.2.6) Datasets

Because the primary scope of this study was the technical validation and assessment of the effectiveness of the developed tool, we deliberately used pre-existing parameters, criteria, and queries extracted from already published reviews, rather than user-defined formulations.

This methodological decision was essential to guarantee the existence of a solid external gold standard against which the algorithm's performance could be objectively compared. Consequently, these input data assumed the role of consolidated external elements and were exclusively channelled to the execution of the preparation scripts.

To ensure the robustness of the validation and the generalisability of the results obtained by the pipeline, we based the selection of systematic reviews on three main axes: thematic diversity, statistical variability, and methodological complexity. The choice of these three reviews aimed to prevent the tool from being optimised for only one type of clinical question, ensuring that the classifier responded consistently across different distributions of positive prevalence and levels of semantic ambiguity. Table 1 summarises the structural and volumetric data of these reviews.

First, we sought to cover distinct sub-areas of biomedical knowledge, namely metabolic meta-analysis, immunology/haematology, and digital health. As an emergent property of this methodological selection, we observed significant statistical variability in the prevalence of positive articles across the datasets. This dispersion allowed us to validate the generalisation capacity of the language models across different technical vocabularies, abstract structures, heterogeneous clinical contexts, and varying conditions of information density. The prevalence of positives in these datasets defined as the proportion of articles marked as eligible (positives) relative to the total number of screened articles in the corpus – reflected the information density of each review. Whereas RV2 had a high prevalence (~17%), testing the classifier's precision in a higher-density environment, RV1 and RV3 represented the realistic large-scale screening scenario with low prevalence (between 0.6% and 1.7%), where the system acted predominantly as an exclusion filter.

Additionally, the inclusion of RV1 (published in 2025) was particularly relevant because its corpus is later than the training cut-off date of most tested models, thereby minimising the risk of data leakage and ensuring that performance reflected rrent logical reasoning ability. Finally, we selected RV3 because of its broad scope and less restrictive inclusion criteria concerning the cost-effectiveness of generic digital technologies. This imposed an added challenge to the LLM's ability to interpret subjectivity, in clear opposition to the higher clinical specificity found in RV1 and RV2.

2.2.7) Prompt Architectures Tested

To optimize the screening tool, we evaluated two distinct prompt architectures. Mode V1, aligned with the method of Li et al. (2024)[2], compiled the three TXT artifacts (instruction, criteria, and document record) into a single user message, preserving the strict sequential order: instruction, followed by criteria, title, and abstract. Mode

V2, inspired by Dennstädt et al. (2024)[3], isolated the instruction and criteria into a dedicated system message, sending only the title/abstract pair as the user message. This second approach followed the idiomatic structure for modern LLMs and allowed leveraging prompt caching mechanisms in cloud APIs, significantly amortising the costs associated with processing successive abstracts. Both architectures enforced a strict output of one affirmative or negative line per evaluated criterion, later consolidated by a binary parser. The experimental matrix (Table 2) crossed these two architectures with the three validation datasets and the two production models.

2.2.8) Experimental Design and Reproducible Configuration

The experimental design crossed three systematic reviews, two prompt architectures (V1 and V2, described in Section 2.2.4), and two production models (DeepSeek-V4-Flash and qwen3:8B). This methodological matrix resulted in twelve independent and complete runs for the scenarios based on DeepSeek-V4-Flash and qwen3:8B. Table 2 summarises the experimental matrix.

Additionally, we ran GPT-4o in V1 mode on RV3 as a contextual baseline (13th run), which was not part of the main 12-run matrix(see supplementary material Figure 1).

During the planning and implementation of the information architecture, we documented several critical technical particularities to ensure exact reproducibility by third parties. The mandatory suppression of the models' internal reasoning mode proved vital, because its latent activation would have made computational time prohibitively long. In the local environment associated with qwen3:8B, this inhibition required directly manipulating the analytical parameters in the native Ollama API (options = {"temperature": 0, "num_predict": 100}), thereby circumventing documented argument propagation failures. In the cloud ecosystem for DeepSeek, we successfully neutralised the reasoning layer by inserting the directive extra_body = {"thinking": {"type": "disabled"}} into the API calls. Concerning the regular expression engine used in the analytical process, we observed an isolated omission of a marginal number of document records (less than 0.1% of the total) whose abstract bodies unintentionally contained the exact structural marking words employed by the system, a limitation that we duly mapped and that had no impact on the overall statistical bias.

2.3 UI Web APP Implementation:

To make the pipeline usable by users without programming experience and to operationalise the human-in-the-loop principle, we wrapped the command-line workflow in a local web application (Django 4.2 LTS, Python 3.9). The application runs entirely on the user's machine as a single-user tool, with a SQLite database and no authentication, accessed via a local browser address. The web application does not duplicate screening logic. Instead, the core of the pipeline (prompt construction for V1/V2, the four provider backends with reasoning suppression, the strict yes/no parser, and PubMed extraction routines) was refactored into a single shared module imported by both the command-line scripts and the web application. This guarantees identical decisions for the same configuration. The data model comprises four entities: Project (review definition), Record (title, abstract, metadata), ScreeningRun (model execution over the corpus), and Prediction (model decision plus optional user override). Exports use the user override when present.The user workflow follows five steps: (1) populate a corpus via PubMed query or file upload; (2) configure eligibility criteria and the model; (3) launch screening (background thread with live progress bar and cancellation support; no external task queues); (4) inspect the list, confirm or correct each decision; (5) export the confirmed inclusion set to CSV/Excel with DOIs

To demonstrate the application end to end, a 20 abstract sample drawn from the RV3 corpus (digital health cost effectiveness) was screened through the interface using DeepSeek-V4-Flash with the V2 prompt structure and deterministic parameters (temperature 0). The corpus was loaded, the run was launched from the browser, and the background worker processed the 20 records in approximately 20 seconds while the progress bar and the partial results updated in real time. The run completed with four articles marked for inclusion by the model and a total consumption of 18,446 tokens. On the review screen the reviewer override was exercised by reclassifying records, and the final inclusion count updated immediately; the confirmed set could then be exported to CSV or Excel with the corresponding metadata.

To confirm that the interactive interface reproduces the established command line pipeline rather than a separate implementation, the same 20 abstracts were independently screened with the standalone command line screener under

an identical configuration. The two paths produced identical decisions for all 20 records (100 % agreement, four inclusions in each), with near identical token usage (18,446 versus 18,452), confirming that the shared core module guarantees consistency between an interactive run and a batch run. The application was further validated across the dimensions summarized in Table 3.

Table 1: Structural and volumetric characterisation of the datasets from the three clinical reviews selected for pipeline validation.

Review	Original paper	Type	Original	Accessed	Filtered T&A	Included
RV1	Mousavi et al. 2025	Meta-analysis	2861	2840	202	29
RV2	Jacobs & Booth 2022	Systematic review	160	131	43	22
RV3	Gentili et al. 2022	Systematic review	1662	1597	81	27

Table 2: Experimental matrix: 3 reviews $\times$ 2 prompt architectures $\times$ 2 production models = 12 runs.

Model	RV1 (Mousavi)	RV2 (Jacobs & Booth)	RV3 (Gentili)
DeepSeek-V4-Flash	V1 + V2	V1 + V2	V1 + V2
qwen3:8B	V1 + V2	V1 + V2	V1 + V2

3. Results

3.1 Pipeline execution

The pipeline successfully completed the twelve planned runs without recording any parsing failures; all computational processing converted 100% of the generated responses into strictly valid binary decisions. The execution of the DeepSeek-V4-Flash model, performed without the active reasoning mode, demonstrated high processing speed, averaging 1.4 seconds per abstract. This efficiency translated into a total time of approximately 40 minutes to

scrutinize the largest corpus (RV1, comprising 2,840 processed records and consuming about 3 million tokens).

In contrast, the Qwen3:8b model deployed in a local environment and bottlenecked by the memory constraints of a 6 GB NVIDIA RTX 2060 graphics card processed each document at a rate of approximately 5 seconds. Consequently, a full execution of RV1 within this local ecosystem required about 3 hours, while the execution of RV3 (version 1) took 1 hour and 40 minutes. The baseline execution using the GPT-4o model on the 1,597 abstracts of RV3 was completed in approximately 50 minutes.From the perspectives of financial viability and workflow sustainability, the disparities between the architectures are substantial. It is estimated that processing the complete execution matrix with GPT-4o would have incurred costs ranging from 40 to 50 USD(see supplementary material). Conversely, the DeepSeek-V4-Flash model executed the exact same volume of API calls for a negligible fraction of that amount, resulting in a total cost of only 0.80 USD. The local Qwen3 solution, meanwhile, operated without any direct monetary costs, albeit imposing a significant temporal burden inherent to local computation

3.2 Overall Model Results

3.2.1) RV1 [7]:

The evaluation of the first review (RV1) focused on a document corpus of 2,840 abstracts. Within this bibliographic un universe, 17 positive cases were identified in the reference standard, corresponding to the recoverable fraction of the 202 articles that had successfully passed the title and abstract (T&A) screening phase in the original meta-analysis. The F1 metric calculation was defined against this canonical denominator of 202 articles. In this analytical context, the Qwen3 model outperformed DeepSeek on the statistical indicator invariant to the denominator, recording the two highest values for Cohen's κ coefficient ($\approx$0.47) and demonstrating conservative behavior by generating a lower proportion of false positives for each captured true positive. The F1 value calculated strictly at the T&A screening level oscillated within a narrow margin between 0.056 and 0.095.

In the cross-architecture analysis, the V2 version of DeepSeek stood out by achieving the highest precision (42.9%), which was accompanied by the inadvertent exclusion of 11 out of the 17 reference standard articles. In the quantification of false negatives, the DeepSeek V1 and Qwen3 V2 runs excluded 6 articles from the reference standard; Qwen3 V1 excluded 9 articles; and DeepSeek V2 inadvertently excluded 11 articles, recording the highest absolute proportion of evidence loss across the entire experimental matrix.Despite the limitations of the reference standard, the Negative Predictive Value (NPV) remained uniformly high across all four runs, ranging from 99.6% to 99.8% Additionally, the workload reduction metrics achieved in this review stabilized between 97% and 99%.

3.2.2) RV2[8]:

A concentrated corpus of 131 evaluated abstracts was processed in RV2, characterized by a positive case prevalence of approximately 17%, corresponding to a final reference standard (*gold standard*) of 22 included articles. The published T&A threshold attested to the selection of 43 documents in the primary screening of the original review, serving as the denominator for the F1 metric calculation. In this high information density scenario, the four tested models recorded F1 values ranging between 0.276 and 0.311, simultaneously achieving T&A sensitivity rates between 39% and 49%. Cohen's κ agreement coefficient

stabilized uniformly within a band between 0.09 and 0.13. Regarding the Negative Predictive Value (NPV), the recorded values amounted to 96.6% for the DeepSeek V1 execution and 90.2% for Qwen3 V2 (Figure 7). Relative to the false negatives against the 22 included articles, the DeepSeek V1 and V2 architectures recorded a single undetected document (FN = 1), while the local Qwen3 model excluded two articles in the V1 configuration and five articles in the V2 configuration Finally, the workload reduction metrics stabilized within a realistic and functional range between 22% and 39%.

3.2.3) RV3[9] :

The evaluation of the third review (RV3) was based on a document corpus of 1,597 abstracts, with a final reference standard established at 35 articles. The T&A screening threshold was set at 81 documents, with this value assuming the role of denominator for the F1 metric. In terms of sensitivity against the final standard, all production runs recorded an excellent performance, with data capture rates ranging between 96.3% and 100%. Specifically, both architectures tested with the DeepSeek model and the V1 version of Qwen3 identified all originally included articles (False Negatives, FN = 0), with the loss of only a single reference document (FN = 1) noted in the Qwen3 V2 execution. NPV was 100% for both DeepSeek configurations and for qwen3 V1, and 99.9% for qwen3 V2, confirming the pipeline's reliability as an exclusion screener.

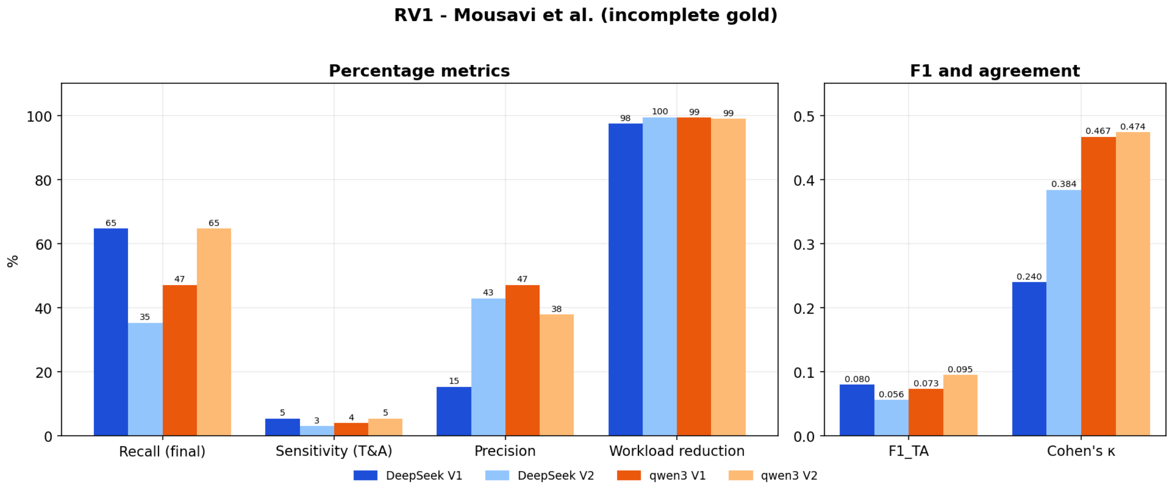


Figure 2 – RV1 (Mousavi et al.): metrics across the four runs (left: percentages; right: F1_TA and Cohen's κ). The high workload reduction (97–100%) is inflated by the very low prevalence and the incomplete reference standard.

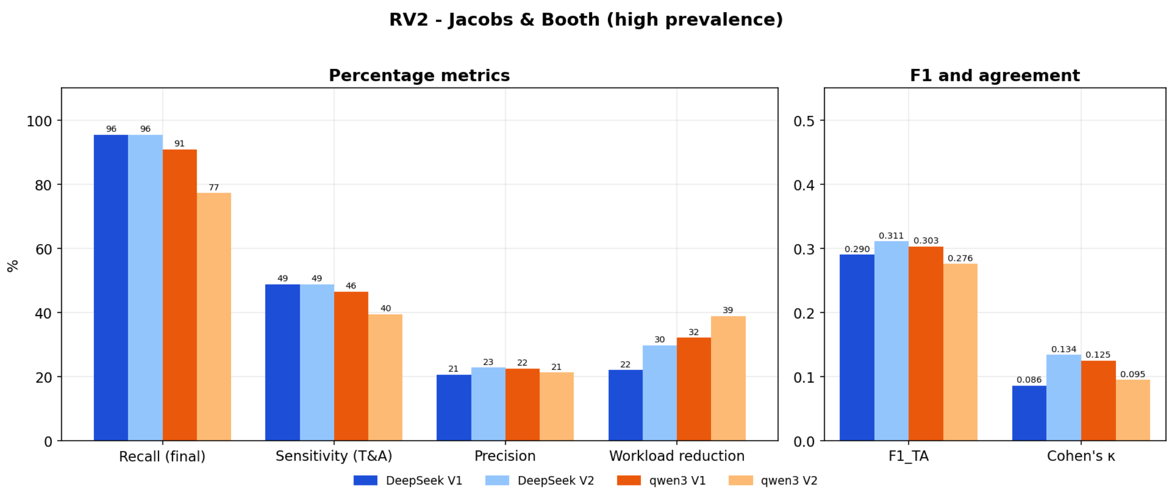


Figure 3 – RV2 (Jacobs & Booth): metrics across the four runs. The highest F1 of the study (0.28–0.31), due to the high prevalence; workload reduction is correspondingly lower.

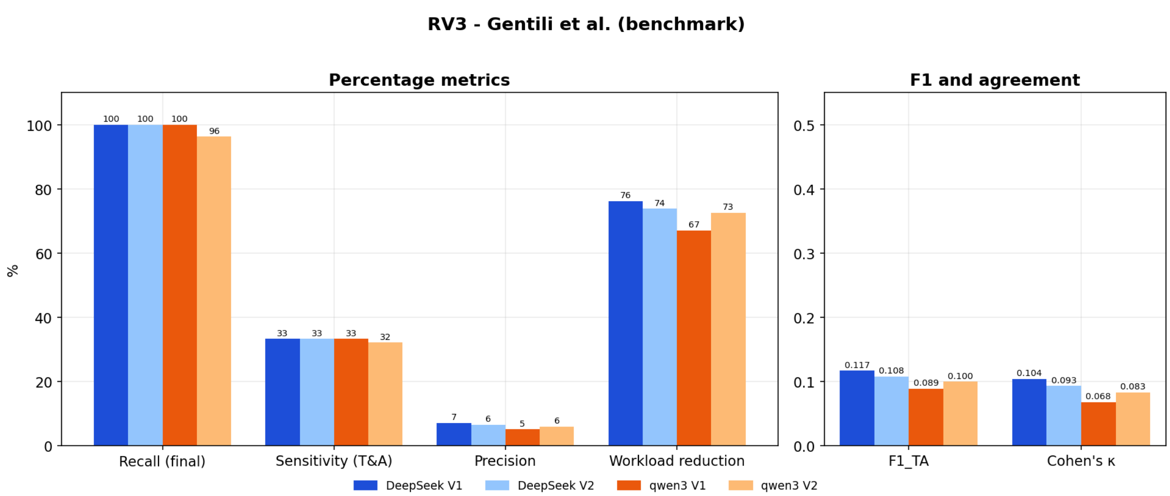


Figure 4 – RV3 (Gentili): metrics across the four runs. The lower-cost models show identical sensitivity but a higher false-positive rate.

3.3 Global Comparisons

3.3.1) F1 Performance at the T&A Screening Level and Global Agreement

In Figure 5 we compare the F1_TA of the production executions (DeepSeek and Qwen3) grouped by review, while Figure 6 illustrates Cohen's κ agreement coefficient and Figure 7 the negative predictive value (NPV). Analyzing the F1 metrics across the datasets, RV2 globally exhibits the highest absolute values of the study, which is a direct artifact of its high statistical prevalence; this same prevalence, however, produces the opposite effect on the NPV, where RV2 records the lowest values of the study (down to 90.2 %), since in a corpus with many true positives a negative prediction is more likely to be erroneous. Conversely, the results for RV1 are skewed by the incompleteness of its reference standard (gold standard), which significantly depresses the apparent F1 and κ metrics in this review regardless of the model used; the NPV, by contrast, remains very high in both RV1 and RV3 (approximately 99.6 to 100 %), as it is dominated by the large number of true negatives inherent to these low-prevalence corpora, confirming the reliability of the pipeline as an exclusion triager.

3.3.2) Structural Versions: V1 vs. V2

Analyzing the average performance of the two lower-cost models, the V1 and V2 instruction versions proved to be virtually indistinguishable in F1_TA across the three evaluated reviews (Figure 8). This empirical finding suggests that the impact of minor adjustments in prompt structure is of second order, being extensively superseded by the intrinsic architectural capacity of the utilized model and the clinical complexity of the domain under analysis.

3.4 Web Application: Functional Demonstration and Validation

The pipeline was delivered as a working local web application that operationalizes the human-in-the-loop workflow described in the methodology. Figure 9 shows the four core screens: the project list (A), the project setup where the eligibility criteria, the corpus (by file upload or PubMed query) and the model are defined (B), the background screening with a live progress indicator (C), and the human review screen, where each decision can be confirmed or overridden before export (D). The application also passed all Django system checks, migration tests, and end-to-end page tests (Table 3).

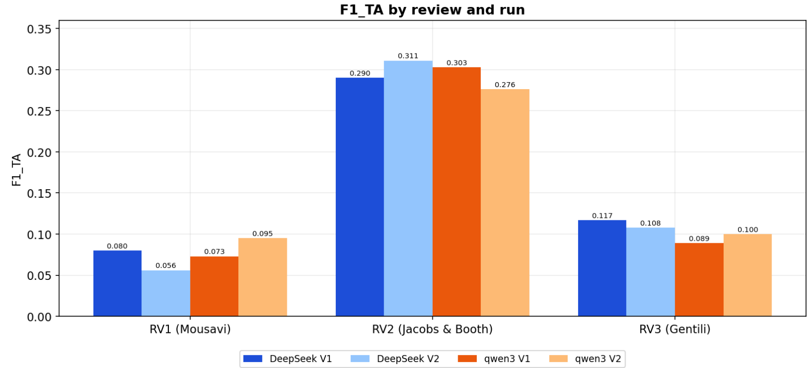


Figure 5 – F1_TA by review and run for the DeepSeek and Qwen3 models. RV2 shows the highest values, due to its higher prevalence.

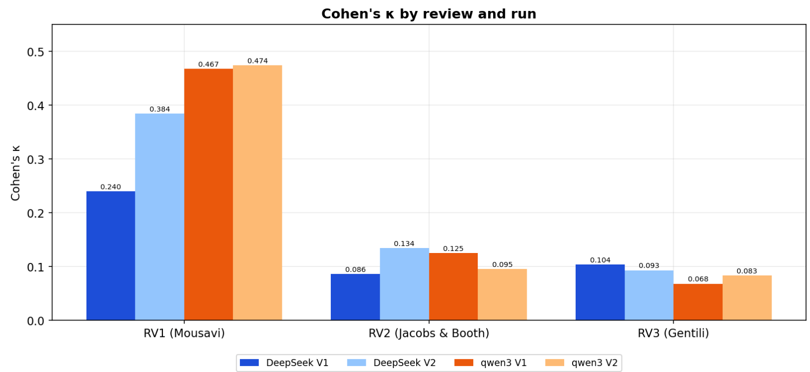


Figure 6 – Cohen's κ coefficient across the twelve production runs grouped by systematic review.

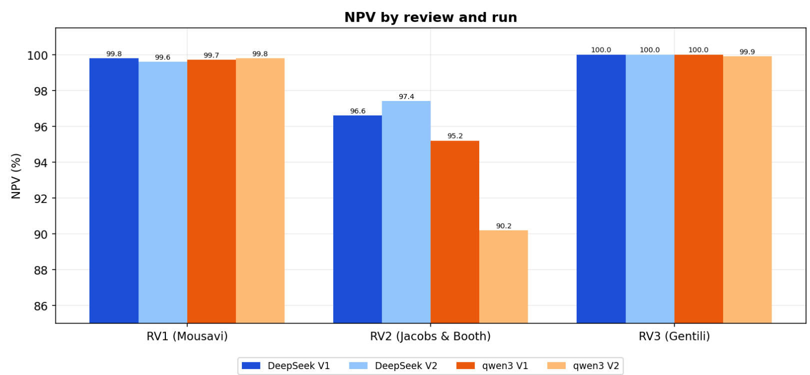


Figure 7 – Negative predictive value (NPV) across the twelve production runs grouped by systematic review.

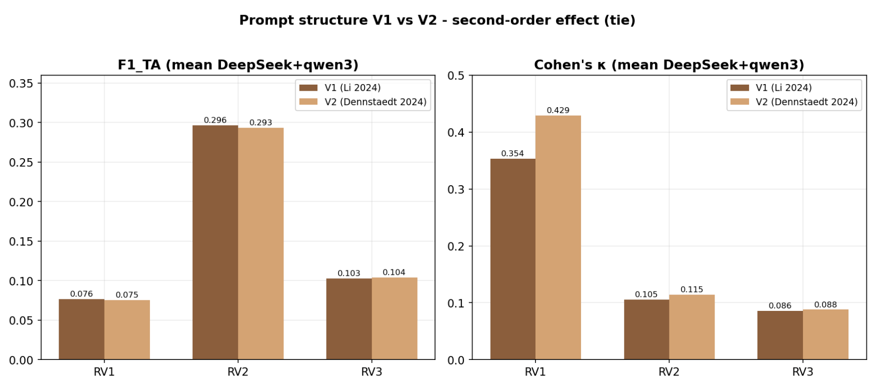


Figure 8 – Prompt structure V1 vs. V2 (mean DeepSeek+Qwen3): nearly equal F1_TA confirms equivalent performance.

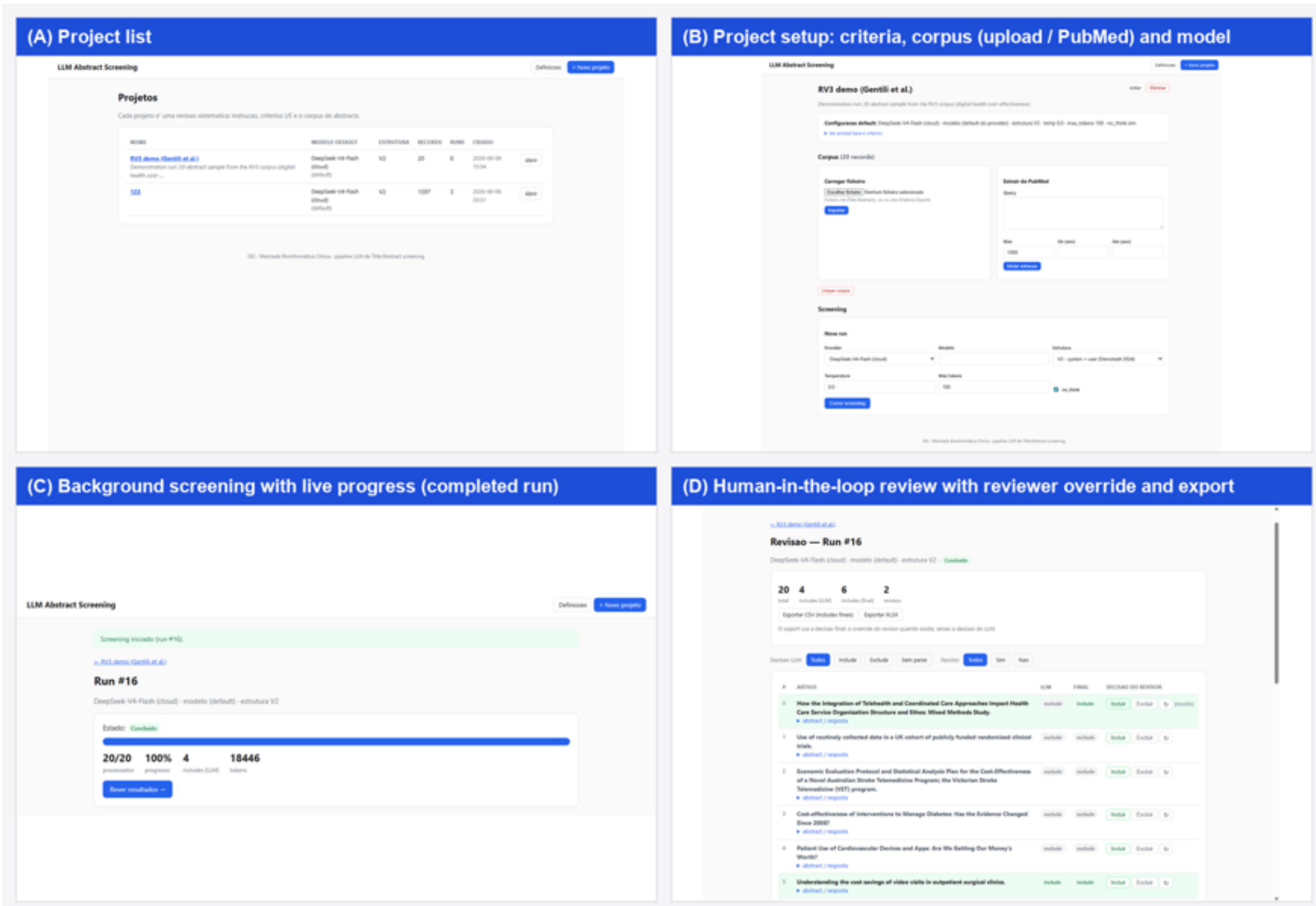


Figure 9 – The four core screens of the web application: (A) project list; (B) project setup (criteria, corpus, model); (C) background screening with live progress; (D) human-in-the-loop review with User override and export.

Table 3: Validation of the web application.

Aspect tested	Method	Result
Configuration integrity	Django system checks	Pass (0 issues)
Database schema	Migration on a clean database	Pass
Page rendering	Every route (list, project, progress, review, export, settings)	Pass (HTTP 200)
Background execution	End-to-end run of a 20-abstract sample	Pass (20/20 in ∼20 s, live progress)
Cloud provider integration	Live DeepSeek-V4-Flash call	Pass (valid binary decisions, 18,446 tokens)
Equivalence with the pipeline	Same 20 abstracts via the independent command-line screener	Pass (20/20 identical, 100% agreement)
Human review and export	User override and CSV / XLSX export	Pass

4. Discussion

4.1 Interpretation of Results (RV1, RV2, and RV3)

4.1.1) RV1

The results of the first review (RV1) must be interpreted with caution, as its gold standard is incomplete. The uniformly low F1 score at the title and abstract (T&A) screening level does not directly reflect poor inferential performance but rather a structural limitation of the dataset. Because the corpus exported exclusively via PubMed contained only 17 of the 202 articles that passed screening in the original review, the remaining literature residing in databases such as EMBASE, the Cochrane Library, or ClinicalTrials.gov, inaccessible through the reproduced query the system's sensitivity faces an insurmountable theoretical ceiling of 8.4%.

Likewise, the unusually high workload reduction (97% to 99%) reported for this review is explained by a combination of logistical and statistical factors and should not be interpreted in isolation as proof of the model's effectiveness. Due to the dispersion of articles across the aforementioned databases, the prevalence of positive articles actually identifiable in the corpus was only 0.6% (17 out of 2840), in stark contrast to the denser prevalence observed in other domains. Given such a low density of relevant articles, any minimally calibrated computational screening model will produce drastic reductions in human reading load. These results demonstrate that when the corpus does not include articles from all the original databases, the metrics reflect this data flaw rather than the model's true capability, reinforcing that the evaluation of the tool should not rely solely on these numerical indicators.

4.1.2) RV2

The evaluation of the indicators obtained from the second review (RV2) demonstrates the direct impact of a high factual prevalence ($\approx$17%) on classifier behaviour. The numerical compression observed in Cohen's κ, which settled uniformly between 0.09 and 0.13, should not be interpreted as an intrinsic failure of semantic alignment. This phenomenon stems from a mathematical limitation inherent to the coefficient under high-prevalence distributions, where the probability of chance agreement is substantially inflated. In parallel, the contraction of the Negative Predictive Value (NPV) to levels such as 96.6% in DeepSeek V1 and 90.2% in Qwen3 V2 reflects this informational density, since a more restricted set of true negatives amplifies the percentage impact of any undue exclusion (false negative), statistically penalising the models.

Furthermore, the modest values achieved for workload reduction (22% to 39%) are the methodologically expected result of conservative behaviour by the language models. In datasets with such a high density of true positives, any algorithm focused on not losing clinical evidence inevitably routes a large proportion of abstracts to manual review. Since the threshold for safely excluding articles is very narrow, the high retention of documents confirms the methodological robustness of the system: the tool adapts its exclusion rate to the density of the review, safeguarding biomedical evidence.

4.1.3 RV3

The analysis of the indicators extracted from the third review (RV3) unequivocally exposes the compromises inherent in the methodological transition between proprietary frontier language models and solutions optimised for low cost. It should be noted that the F1 score obtained for T&A screening must be interpreted as a lower bound. This is because, as a precaution, the metric calculation assumes no overlap between the false positives generated by the model and the 54 articles from the original meta-analysis that were not included in our corpus. Since there are likely relevant articles in that blind spot, the system's true performance will, by definition, be higher than the minimum value reported here. Among the low-cost models, DeepSeek showed a marginal superiority over Qwen3, reversing the performance pattern recorded in RV1. This fluctuation suggests that the effectiveness of each architecture is, to a large extent, conditioned by the specific characteristics of each review's topic.

4.2 Cross-cutting Patterns, Prompt Effect, and Cost–Quality Trade-off.

From the cross-sectional analysis of the complete matrix, clear behavioral trends emerge that warrant detailed examination.

Recall against the final included articles is high and stable in reviews with a reliable gold standard (RV2: 77–95.5%; RV3: 96–100%). The marked drop in RV1 (35–65%) stems exclusively from the documented incompleteness of its gold standard, not from model failure. In RV2 and RV3, the loss of relevant articles (false negatives) is residual: 0–5 documents missed out of 22–27 included. In the biased setting of RV1, however, the models inadvertently excluded 6–11 of the 17 articles present in the corpus, the most serious systemic failure, which strongly reinforces the necessity of the *human-in-the-loop* paradigm as an indispensable methodological safeguard.

The initial hypothesis that the V2 structure (system + user prompt, idiomatic for conversational models) would be systematically superior was not confirmed. Across the six within-model, within-review comparisons, V1 performed better in three cases (DeepSeek-RV1, DeepSeek-RV3, qwen3-RV2) and V2 in the other three (DeepSeek-RV2, qwen3-RV1, qwen3-RV3). This perfect split shows that, when both structures are well formed, the isolated effect of prompt engineering is marginal and inconsistent, subordinate to the model's intrinsic capability and the nature of the domain. The practical implication is reassuring: either architecture guarantees inferential robustness, removing the need for exhaustive prompt micro-optimisation.

Domain complexity as the dominant factor. The difficulty of the domain is the primary driver of performance. RV3, with broad eligibility criteria "cost-effectiveness of any digital technology), yields κ values of only 0.07–0.10 for the production models. In contrast, RV1 and RV2, anchored in more restrictive and objective clinical criteria, achieve κ around 0.47. This pattern is in full agreement with Adam et al. (2026)[1]: variance across thematic contexts comfortably outweighs the marginal gains from technical configuration adjustments.

Cost–quality trade-off. Replacing GPT-4o with cheaper models (DeepSeek-V4-Flash cloud and local qwen3:8B) kept the total expenditure under 10 USD, but entailed a measurable cost in precision, particularly in the diffuse domain of RV3 (see Supplementary Materials for the detailed comparison with GPT-4o). Among the low-cost models, the local qwen3 outperformed DeepSeek in RV1, tied in RV2, and fell slightly behind in RV3, with no statistically conclusive differences. The practical heuristic is: (a) for well-delineated domains, economical models (local or cloud) are sufficient; (b) for ambiguous criteria, a frontier model may be justified, but scaling should be decided on a case-by-case basis, guided by the NPV obtained during validation; (c) the local model additionally offers privacy advantages and long-term reproducibility.

4.3 Contextualising F1 Performance

The F1 score consistently ranged between 0.06 and 0.31. Although this range might appear, at first glance, to indicate unsatisfactory performance, it constitutes a statistically expected result for four fundamental reasons. From a mathematical standpoint, prevalence dictates the theoretical limits of the metric: the lower the prevalence of positives, the lower the achievable precision, and consequently, the lower the F1 score. At the T&A (Title and Abstract) level, the prevalence is approximately 5% in RV3 and 7% in RV1, rising to about 33% in RV2, precisely the review where the F1 score reaches its peak (0.31). Since the F1 score is the harmonic mean of precision and sensitivity, in screening scenarios with low prevalence (typical of real-world systematic reviews), this metric inevitably tends to collapse.

Secondly, compared to recently published benchmarks, Adam et al. (2026)[1] report a median F1 score of 0.276 $\pm$ 0.172 across 515 prompts. The values obtained in the present work fall within one standard deviation of that median, demonstrating total consistency with the current behavior of frontier models under identical operational conditions.

Conceptually, the F1 score assumes a symmetric cost between false positives and false negatives, which is an invalid premise in medical bibliographic screening. A false negative translates to the irreparable loss of relevant clinical evidence; in contrast, a false positive represents merely a marginal increase of seconds in the human review workload. Consequently, metrics such as Negative Predictive Value (NPV) and Workload Reduction assume superior methodological adequacy in this use case. Finally, due to the technical issues already outlined (Section 4.1), the reported F1 score, particularly in RV3, acts strictly as an underestimated lower bound.

4.4 Contribution and Positioning in the Literature

Our results align closely with published benchmarks. Adam et al. (2026)[1] report a median recall of 83.6% $\pm$ 17%; the sensitivity of our pipeline in RV2 and RV3 (77–100%) falls within that upper range. López-Pineda et al. (2025) [10] concluded that LLMs are "useful for exclusion, but inclusion requires a human" (NPV $\geq$96%); the NPV we obtained ($\geq$99% in most runs, $\geq$90% in RV2) independently corroborates this. The tested models do not reach the human agreement level (maximum $\kappa = 0.47$), and as Oami et al. (2026) [11] note, current limitations preclude full automation the LLM acts as a primary filter, with final decisions remaining under human supervision.

Two aspects distinguish this study from most previous work. First, rather than evaluating isolated configurations, we present a comprehensive 12-run matrix that separates the effects of prompt, model, and domain. Second, the direct comparison among a local open-weights model, a low-cost cloud model, and a frontier model on consumer hardware establishes a useful precedent for research teams with limited AI funding.

4.5 Limitations and Future Perspectives The first limitation stems from incomplete reference standards. In RV1, only 17 positives were flagged from an expected universe of 202 validated articles, as part of the corpus resided in repositories unreachable through strict PubMed query reproduction, artificially depressing sensitivity and F1 and rendering these metrics strictly exploratory. In RV3, only 27 of 34 validated articles were captured, imposing a theoretical recall ceiling of 79.4%, despite the pipeline achieving 100% recall on the effectively cross-referenced subset. Both cases require manual recovery of missing literature from peripheral databases (EMBASE, Cochrane) for metrics to transition to absolute interpretable values.

The second limitation is the absence of stochastic replication. Each configuration ran on a single execution due to computational constraints. Although T=0 maximises determinism, the lack of multiple iterations precluded rigorous confidence interval calculation. Future work plans 3 to 5 runs per configuration with 95% bootstrap interval modelling over F1_TA.

A parser bias was also detected, causing minor document loss (21 to 65 records per review) attributable to regular expression limits (DOTALL edge-case), assumed to have non-zero but unsystematic statistical impact across inclusion classes.

Finally, the binary decision architecture precludes ROC curve construction and fine threshold optimization. Future work contemplates evolution towards a Likert-scale probabilistic classification (1 to 5) with threshold sweep, estimated to yield approximately 5 F1-point gains (Dennstädt, 2024)[3], complemented by multi-model voting algorithms (Li, 2024)[2] aggregating DeepSeek, Qwen3 and a third open-architecture model, anticipated to raise the precision upper bound in high-ambiguity scenarios.

5. Conclusion

This study validates a minimalist, open-source, single-LLM pipeline running on consumer hardware as a viable and economical tool for semi-automated title and abstract screening in systematic reviews. Across three published reviews, the pipeline achieved high recall, negative predictive values $\geq 99\%$, and substantial reductions in manual workload, at negligible cloud cost and zero local cost. Prompt structure had no effect on performance; domain complexity was the dominant factor. The local qwen3:8B offers complete data privacy and zero API dependency. Human-in-the-loop validation of exclusions remains indispensable. Future work will address incomplete gold standards, single-run design, and binary decisions through database recovery, stochastic replication, and probabilistic multi-model classification. In summary, this open-source proof of concept provides a robust, accessible, and secure alternative for optimising evidence synthesis.

Authors' Contributions

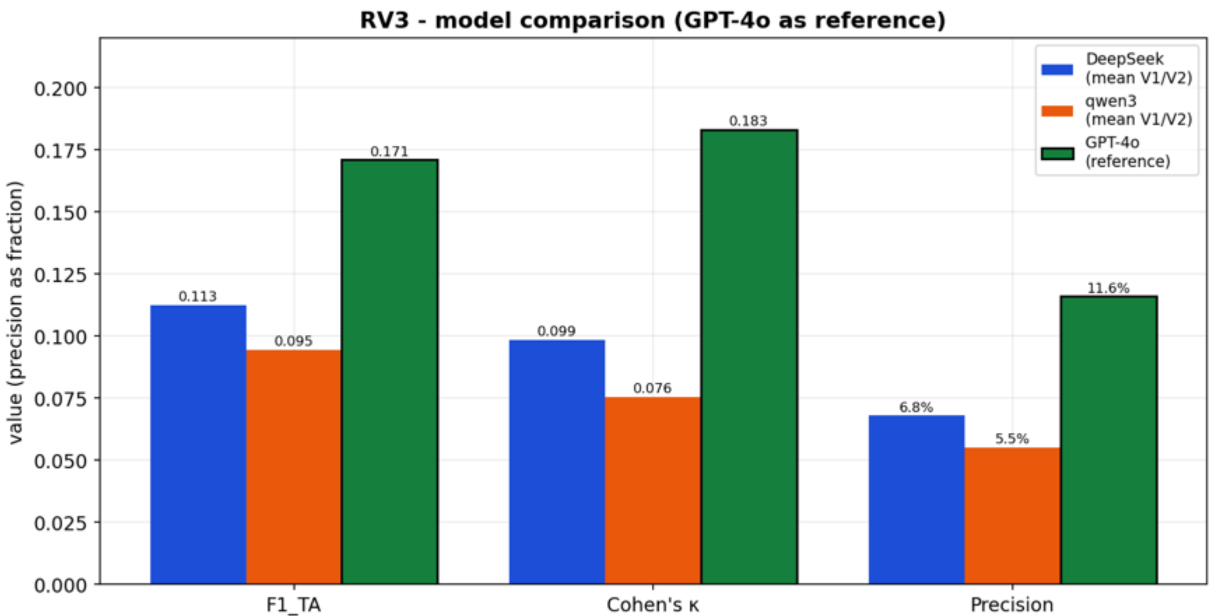
Miguel Alves was responsible for the conceptualisation of the study, the design of the pipeline architecture, the overall implementation, and the writing of the manuscript. **Gonçalo Bernardo** developed the entire codebase, implemented the screening pipeline and the local web application, and contributed to the manuscript. **Júlia Gonçalves, Alexandra Rodrigues**, and **André Oliveira** contributed substantially to the writing of the manuscript, the curation of the review datasets, and the elaboration of the systematic reviews used for validation. All authors critically reviewed and approved the final version of the manuscript.

Conflicts of interest

The authors declare no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary Material:

In the specific domain evaluated across all three architectures (RV3), the frontier model (GPT-4o) is introduced as a high-performance baseline. As demonstrated in Supplementary Figure 1, GPT-4o exhibits a clear superiority in Precision, F1_TA, and κ metrics, practically doubling the agreement coefficient of the best open-architecture model on this dataset. This discrepancy directly quantifies the efficiency trade-off incurred when transitioning from a state-of-the-art frontier model to more affordable or locally deployed alternatives. Among the lower-cost options, DeepSeek marginally outperforms Qwen3 in RV3, although an inversion of this performance hierarchy is observed under the specific conditions of RV1.



Supplementary Figure 1 – RV3: model comparison with GPT-4o as a baseline reference. V1/V2 averages for DeepSeek and Qwen3 are plotted against GPT-4o to highlight the performance gap across Precision, F1_TA, and κ.

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